AMAZON BRAZIL

SQL CASE STUDY

Marketplace Performance & Strategic Analysis

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LATIN AMERICA'S LARGEST E-COMMERCE MARKET

ORDERS • SELLERS • CUSTOMERS • PAYMENTS • REVIEWS

EXECUTIVE SUMMARY

Amazon Brazil is scaling rapidly in Latin America's largest e-commerce market. The operations and strategy team needs a clearer picture of how the platform is performing — This project uses SQL to analyze that data and understand marketplace performance, customer behavior, seller activity, revenue, and operations..

CUSTOMERS

Which customers are driving revenue?

Top spender, order frequency by state, and buyers who never review.

SELLERS

Which sellers are delivering?

Order volume by seller state, dormant accounts, and above-average sellers.

CATEGORIES

Which categories are growing?

Cumulative revenue, top 5 categories, review scores, and never-ordered products.

FULFILLMENT

Where is fulfillment breaking down?

Median delivery time, regional satisfaction, payment mix, and peak demand month.

Goal: use SQL to turn raw transactional data into decisions.

DATASET OVERVIEW

orders
order_id
customer_id
order_status
order_purchase_timestamp
order_approved_at
order_delivered_carrier_date
order_delivered_customer_date
order_estimated_delivery_date

customers
customer_id
customer_unique_id
customer_zip_code_prefix
customer_city
customer_state

sellers
seller_id
seller_zip_code_prefix
seller_city
seller_state

order_items
order_id
order_item_id
product_id
seller_id
shipping_limit_date
price
freight_value

order_payments
order_id
payment_sequential
payment_type
payment_installments
payment_value

geolocation
geolocation_zip_code_prefix
geolocation_lat
geolocation_lng
geolocation_city
geolocation_state

products
product_id
product_category_name
product_name_lenght
product_description_lenght
product_photos_qty
product_weight_g
product_length_cm
product_height_cm
product_width_cm

order_reviews
review_id
order_id
review_score
review_comment_title
review_comment_message
review_creation_date
review_answer_timestamp

product_category_name_translation
product_category_name
product_category_name_english

leads_closed
mql_id
seller_id
sdr_id
sr_id
won_date
business_segment
lead_type
lead_behaviour_profile
has_company
has_gtin
average_stock
business_type
declared_product_catalog_size
declared_monthly_revenue

leads_qualified
mql_id
first_contact_date
landing_page_id
origin

11 tables • 70 columns
Orange = primary key

BUSINESS OBJECTIVES

01	**Plan Regional Fulfillment**	Logistics: identify states with the highest seller-side order volume to support regional fulfillment planning.
02	**Grow Revenue by Category**	Growth & Merchandising: track category revenue growth and identify categories for promotional focus.
03	**Build Customer Value**	CRM & Regional Marketing: identify high-value customers, compare order frequency by state, and improve review engagement.
04	**Raise Seller Quality**	Seller Growth & Success: identify dormant sellers and recognize sellers performing above the platform average.
05	**Improve Experience & Plan Demand**	Finance, CX, Catalogue & Forecasting: analyze payment behaviour, customer satisfaction, inactive products, and peak demand.

ANALYTICAL APPROACH

CUSTOMER & MARKET INTELLIGENCE

- High-value customer identification
- Customer orders by state
- Review engagement analysis

SELLER PERFORMANCE & FULFILLMENT

- Orders by seller state
- Dormant seller detection
- Above-average seller performance

REVENUE & PRODUCT ANALYTICS

- Cumulative category revenue
- Top 5 revenue categories
- Products never ordered

OPERATIONS & CUSTOMER EXPERIENCE

- Payment type & order value
- Median delivery time
- Category satisfaction
- State satisfaction
- Peak order month

CUSTOMER & MARKET INTELLIGENCE

Which customers drive revenue, how often they order across states, and who stays silent after purchase.

High-Value Customer	Find the single highest-spending customer for the first premium loyalty tier.
Orders by State	See how order frequency varies by customer location for state-level campaigns.
Review Engagement	Identify buyers who order but never review, to target with reminders.

High-Value Customer Identification

Q1 BUSINESS OBJECTIVE

CRM TEAM

Find the customer who has spent the most money across all their orders.

SQL QUERY

```
select
    t1.customer_unique_id,sum(t3.payment_value) as money_spent
from customers as t1
join orders as t2
    using(customer_id)
join order_payments as t3
    using(order_id)
group by 1
order by money_spent desc limit 1;
```

BUSINESS CONTEXT

- Supports the launch of Amazon Brazil's first premium loyalty tier.
- Identifies the top spender to onboard personally into the program.
- Shows what a top-tier customer profile looks like.

Customer Orders Across States

Q2 BUSINESS OBJECTIVE

REGIONAL MARKETING TEAM

Find the total number of orders placed by each customer, broken down by the state they live in.

SQL QUERY

```
select
    t1.customer_unique_id as customer_unique_id,
    t1.customer_state as customer_state,
    count(t2.order_id) as total_orders
from customers as t1
join orders as t2
    using(customer_id)
group by 1, 2;
```

BUSINESS CONTEXT

- Shows how order frequency varies by customer location.
- Targets low-frequency states with acquisition offers.
- Rewards high-frequency states with loyalty perks.

Customer Engagement Analysis

Q3 BUSINESS OBJECTIVE

POST-PURCHASE ENGAGEMENT TEAM

Identify customers who have placed orders but never left a review.

SQL QUERY

```
select
    customer_unique_id
from customers
join orders
    using(customer_id)
where not EXISTS(
    select 1
    from order_reviews
    where order_reviews.order_id = orders.order_id);
```

BUSINESS CONTEXT

- More reviews improve product discoverability and buyer trust.
- Finds regular buyers who never review.
- Creates the audience for review reminder emails and in-app nudges.

SELLER PERFORMANCE & FULFILLMENT

Where seller-side demand sits, which accounts are dormant, and who beats the platform average.

Orders by Seller State	Show which states generate the most seller-side order volume.
Dormant Sellers	List sellers registered on the platform with zero fulfilled orders.
Above-Average Sellers	Find sellers who fulfilled more orders than the platform average.

Seller State Contribution Analysis

Q4 BUSINESS OBJECTIVE

LOGISTICS TEAM

Find the total number of orders fulfilled by each seller state.

SQL QUERY

```
select
    t1.seller_state, count(distinct t2.order_id) as total_orders
from sellers as t1
join order_items as t2
    using(seller_id)
group by 1
order by total_orders desc;
```

BUSINESS CONTEXT

- Regional fulfillment center investments are being planned.
- Shows which seller states generate the most order volume.
- Sends infrastructure budget to where demand actually is.

Dormant Seller Detection

Q5 BUSINESS OBJECTIVE

SELLER GROWTH TEAM

Identify sellers who registered on the platform but have never fulfilled a single order.

SQL QUERY

```
select
    t1.seller_id,
    t1.seller_city,
    t1.seller_state
from sellers as t1
where not exists(select 1
                 from order_items as t2
                 where t2.seller_id = t1.seller_id);
```

BUSINESS CONTEXT

- Builds the list for quarterly re-engagement campaigns.
- Finds every account with zero fulfilled orders.
- Flags accounts that need a nudge or a review for removal.

Seller Performance Benchmarking

Q6 BUSINESS OBJECTIVE

SELLER SUCCESS TEAM

Find sellers who have fulfilled more orders than the average seller on the platform.

SQL QUERY

```sql
with fulfilled_orders as (
    select
    seller_id, count(DISTINCT order_id) total_orders
    from order_items
    group by 1),
average as (
    select avg(total_orders) as avg_orders
    from fulfilled_orders)
select
    t1.seller_id, t1.total_orders
from fulfilled_orders as t1
cross join average as t2
where t1.total_orders > t2.avg_orders;
```

BUSINESS CONTEXT

- Builds the invite list for the “Top Seller” badge program.
- Only sellers above the platform average qualify.
- Gives a numeric baseline for above-average performance.

REVENUE & PRODUCT ANALYTICS

How category revenue builds, which categories lead, and which listings never sell.

Cumulative Category Revenue	Track the running revenue total in each category across the order timeline.
Top Revenue Categories	Rank the top 5 categories by total revenue for sale-event planning.
Never-Ordered Products	List products with no orders for the quarterly catalogue clean-up.

Category-Level Revenue Growth

Q7 BUSINESS OBJECTIVE

GROWTH TEAM

For each product category, calculate the cumulative revenue generated as orders come in over time.

SQL QUERY

```
with category_revenue as (
    select
        t1.product_category_name_english as product_category_name,
        DATE(t4.order_purchase_timestamp) as order_date,
        sum(t3.price) as revenue
    from product_category_name_translation as t1
    join products as t2
        using(product_category_name)
    join order_items as t3
        using(product_id)
    join orders as t4
        using(order_id)
    group by 1,2)
select
    product_category_name, order_date, revenue,
    round(sum(revenue) over(partition by product_category_name order by order_date
            rows between unbounded preceding and current row), 2) as cumulative_sum
from category_revenue;
```

BUSINESS CONTEXT

- Shows how revenue builds up within each category.
- Spots categories that hit targets early versus late in the month.
- Helps plan flash sales and push notifications.

Top Revenue-Generating Categories

Q8 BUSINESS OBJECTIVE

MERCHANDISING TEAM

Find the top 5 product categories by total revenue.

SQL QUERY

```sql
select
    t1.product_category_name_english as product_category,
    sum(t3.price) as total_revenue
from product_category_name_translation as t1
join products as t2
    using(product_category_name)
join order_items as t3
    using(product_id)
group by 1
order by total_revenue desc limit 5;
```

BUSINESS CONTEXT

- Decides homepage space and promo budget for Black Friday, Dia das Mães and Cyber Monday.
- Total revenue by category is the primary input.
- Highlights the categories to feature first.

Product Demand & Catalogue Clean-up

Q9 BUSINESS OBJECTIVE

CATALOGUE OPERATIONS TEAM

Find all products that have never been ordered.

SQL QUERY

```
select
    product_id
from products
where not EXISTS(
        select 1
        from order_items
        where
            order_items.product_id = products.product_id);
```

BUSINESS CONTEXT

- Dead listings waste search index space and confuse customers.
- They also inflate category browse pages.
- Gives the exact list to review, reprice, or remove.

OPERATIONS & CUSTOMER EXPERIENCE

Payments, delivery speed, satisfaction by category and state, and the peak demand month.

Payment Preference	Compare payment type usage and average order value for provider talks.
Delivery Time	Measure the true middle ground of how long customers wait.
Customer Satisfaction	Review scores by product category and by delivery state.
Peak Demand	Find the historically busiest order month for capacity planning.

Payment Preference Analysis

Q10 BUSINESS OBJECTIVE

FINANCE & PARTNERSHIPS TEAM

Which payment method do customers use the most, and what is the average order value for each payment type?

SQL QUERY

```sql
with payments as (select
    order_id, payment_type, sum(payment_value) as order_value
from order_payments
group by 1, 2)
select
    payment_type, count(DISTINCT order_id) as total_orders,
    avg(order_value) as avg_order_value
from payments
group by payment_type
order by total_orders desc;
```

BUSINESS CONTEXT

- Supports renegotiation with payment providers.
- Shows whether credit card, boleto, voucher or debit card dominates.
- Reveals if high-value orders cluster around one payment type.

Delivery Time Performance

Q11 BUSINESS OBJECTIVE

LOGISTICS DIRECTOR

Find the median delivery time (in days) between order placement and actual delivery.

SQL QUERY

```
with delivery as (select
    DATEDIFF(order_delivered_customer_date, order_purchase_timestamp) as delivery_days,
    ROW_NUMBER()
    over(order by DATEDIFF(order_delivered_customer_date, order_purchase_timestamp))as row_num,
    count(*) over() total_rows
from orders
where order_delivered_customer_date is not null)
select
    round(avg(delivery_days),0) as median_delivery_time
from delivery
where row_num in(
            floor((total_rows + 1) / 2),
            ceil((total_rows + 1) / 2));
```

BUSINESS CONTEXT

- Median shows the typical wait for most customers.
- Avoids skew from extremely delayed remote-interior orders.
- Gives a fair view of delivery performance.

Customer Satisfaction Evaluation

Q12 BUSINESS OBJECTIVE

CUSTOMER EXPERIENCE TEAM

Find the average review score for each product category.

SQL QUERY

```
select
    t1.product_category_name_english as product_category,
    avg(t4.review_score) as avg_review_score
from product_category_name_translation as t1
join products as t2
    using(product_category_name)
join order_items as t3
    using(product_id)
join order_reviews as t4
    using(order_id)
group by 1;
```

BUSINESS CONTEXT

- Low scores often signal fulfillment, quality or listing-mismatch issues.
- Tracks satisfaction at the category level.
- Shows which categories need intervention first.

Regional Customer Satisfaction

Q13 BUSINESS OBJECTIVE

CUSTOMER EXPERIENCE TEAM

Find which Brazilian states have the highest average customer review score for orders delivered there.

SQL QUERY

```
select
    t1.customer_state as state,
    avg(t3.review_score) as avg_customer_review
from customers as t1
join orders as t2
    using(customer_id)
join order_reviews as t3
    using(order_id)
group by 1
order by avg_customer_review desc limit 1;
```

BUSINESS CONTEXT

- Last-mile experience differs across Brazil, e.g. São Paulo and Rio versus the North and Northeast.
- Links delivery destination to satisfaction.
- Helps push logistics partners to improve in weaker states.

Order Demand Trend Analysis

Q14 BUSINESS OBJECTIVE

DEMAND FORECASTING TEAM

Find the month with the highest number of orders placed across the entire platform.

SQL QUERY

```
select
    month(order_purchase_timestamp) as order_month,
    count(order_id) as total_orders
from orders
group by 1
order by total_orders desc limit 1;
```

BUSINESS CONTEXT

- Reveals the historical peak order month.
- Guides warehouse capacity and temporary delivery staffing.
- Sets the timing of campaigns for the next peak season.

RECOMMENDATIONS

01 Invest Where Demand Is

Use seller-state order volume to guide regional fulfillment and infrastructure planning.

02 Back the Top Categories

Give high-revenue categories greater promotional visibility and budget during major sale events.

03 Launch Premium Loyalty

Use the highest-spending customer as a starting point for premium loyalty and retention strategies.

04 Activate Sellers

Re-engage dormant sellers and recognize sellers performing above the platform average.

05 Fix Weak Regions

Use delivery time and state-level review scores to identify areas where logistics performance needs improvement.

06 Grow Reviews & Clean Catalogue

Target customers who never review and review products that have never received an order.

CONCLUSION

01 One Analytical View

Customers, sellers, products, orders, payments and reviews are connected through 14 SQL queries.

02 Decisions, Not Just Data

Each query answers a real question from logistics, growth, finance, CRM, CX or merchandising.

03 Clear Priorities

Demand by state, top categories, key customers and seller activity show where to act first.

04 Next Step

Extend the work into a portfolio case study with documented findings and a GitHub repository.



THANK YOU

SQL-Driven Insights for Smarter Marketplace Decisions

amazon